Field Visit & Market Audit Platform

Comprehensive Full-Lifecycle Engineering, Delivery & Technical Architecture Report

Recipient: General Manager (GM)
Author: Lead Field Platform Developer
Reporting Period: Aug 27, 2026 – Oct 24, 2026 (9 Weeks / 59 Days)
Status: v3.0 Enterprise Operational & Concurrency Hardened

EXECUTIVE OVERVIEW & BUSINESS PURPOSE

This comprehensive report details the complete engineering lifecycle of the **Field Visit & Market Execution Platform** from architectural inception on **August 27, 2026** through enterprise release **v3.0 on October 24, 2026**. Over **9 continuous weeks (59 calendar days / 34 dedicated active development days)**, the platform evolved into a mission-critical operating system: integrating ERP/CUSTMASTER databases across 2,535 commercial outlets, eliminating MSSQL Deadlock 1205 & timeout errors via native T-SQL derived table batching with `WITH (HOLDLOCK)`, standardizing dual-engine failover (Azure SQL Server & MySQL), syncing all 20 canonical supervisors, calibrating Dairy classification denominators, deploying persistent Master File cache invalidation, and enforcing closed-loop GM directive tracking with visual photo proof.

286 Hrs	164	2,535	20 Users	218 Modules	9 Weeks
TOTAL ENGINEERING HOURS	PRODUCTION COMMITS	ACTIVE OUTLETS AUDITED	FIELD SUPERVISORS SYNCED	185,000+ LINES CODE	FULL LIFECYCLE SPAN

1. Full Lifecycle Engineering Effort & Time Allocation

Total: 286 Hours across 7 Phases

Development Phase	Timeline & Span	Primary Engineering Scope & Modules Delivered	Time Spent
Phase 1: Platform Foundation & Offline PWA	Aug 27 – Sep 10, 2026 (Weeks 1–2 · 10 Commits)	Next.js 16 architecture, Cloudinary to local photo storage migration, PWA Service Worker offline caching, background sync engine, client data usage tracking, and initial supervisor visit wizard.	**42 Hours**
Phase 2: Database Migration & Cloud CI/CD	Sep 14 – Sep 23, 2026 (Weeks 3–4 · 11 Commits)	Full SQL Server migration from legacy MySQL, Azure App Service containerized Next.js standalone build pipeline, GitHub Actions CI/CD workflows, runtime Azure environment injection, and initial admin seeding.	**36 Hours**
Phase 3: CUSTMASTER Sync & GM Directives	Sep 24 – Sep 30, 2026 (Weeks 4–5 · 37 Commits)	CUSTMASTER ERP master data integration (Manager, Supervisor, Route, Classification), GM Directive & Supervisor Resolution Proof workflow, camera-only proof enforcement, Sub-Admin role, TT vs MT channel separation, and Saifullah vs Saif disambiguation.	**48 Hours**
Phase 4: Analytics Suite & Lightbox Engine	Oct 01 – Oct 08, 2026 (Week 6 · 14 Commits)	Interactive high-resolution image audit lightbox with zoom, pan, and rotation dock; multi-asset chiller/freezer tracking; initial Visit Execution & Asset Summary Table with Excel export; Power SKU APIs.	**42 Hours**
Phase 5: Executive Overhaul & Hardening	Oct 09 – Oct 14, 2026 (Week 7 · 34 Commits)	Luxury executive dashboard redesign (donut KPIs, high contrast typography), strict Power SKU rule enforcement (MT=24, TT=27), strict directive isolation by photoId, Fit-to-Screen table mode, dual-engine Azure SQL/MySQL, and "All" records export pagination.	**46 Hours**
Phase 6: Master Sync Pipeline & MSSQL MERGE	Oct 14 – Oct 19, 2026 (Week 8 · 15 Commits)	Master files Excel upload, disk persistence & instant memory cache invalidation; dynamic Power SKU serving (24 MT / 27 TT with Scotch Vanilla consolidation); outlet classification visit frequency guidelines (A: 4/mo, B: 2/mo, C: 1/mo, D: 0.5/mo) with achievement tracking; MSSQL native T-SQL MERGE preventing comment loss on deploys; dynamic route-to-channel resolution; and PWA API offline route bypass.	**28 Hours**
Phase 7: Concurrency Hardening, Canonical Auth & Denominators	Oct 20 – Oct 24, 2026 (Week 9 · 43 Commits)	MSSQL Deadlock 1205 & timeout 499 resolution via derived table chunking and `WITH (HOLDLOCK)`; elimination of duplicate parameter `@p0` in multi-statement transactions; directory-wide synchronization of all 20 canonical supervisors with REST API migration; recalibration of Dairy classification denominators (2,535 stores); Cold Chain data-visual alignment (97%/3%); supervisor directive acknowledgement date tracking; Near-Expiry channel exclusion; and Full-Page modal analytics.	**44 Hours**
TOTAL CUMULATIVE ENGINEERING TIME INVESTED ACROSS LIFECYCLE			**286 Hours**

2. Complete Feature Inventory Developed (End-to-End)

Module	Feature Description & Capabilities Built	Business & Leadership Impact	Delivered
High-Concurrency MSSQL Engine	Engineered derived table batching, table locking hints `WITH (HOLDLOCK)`, isolated `sql.Request` transaction scopes, and exponential backoff retry wrappers in `lib/db.ts`.	Eliminates Deadlock 1205 and timeout 499 during simultaneous multi-supervisor upload peaks.	**PRODUCTION** Oct 22, 2026
Directive SLA & Acknowledgement	Date-stamped supervisor acknowledgement tracking on flagged audit directives (`AuditActionsTrackerSummary`), capturing exact viewing timestamps prior to resolution photo upload.	Enforces end-to-end operational accountability and quantifies supervisor response latency.	**PRODUCTION** Oct 22, 2026
Canonical Supervisor Directory	Automated synchronization of all 20 canonical field supervisors, employee codes, and bcrypt credentials matching live `dandyapp.tech`. Migrated supervisor edits to resilient REST APIs.	Eradicates login lockouts, authentication drift, and Server Action timeout crashes.	**PRODUCTION** Oct 24, 2026
Calibrated Master Denominators	Calibrated Dairy and Juice classification denominator hierarchy to exact master counts (2,535 stores: Class A: 341, Class B: 551, Class C: 648, Class D: 434, Class E: 561) in dashboard analytics.	Guarantees 100% mathematically verified tier coverage and penetration rates across sales regions.	**PRODUCTION** Oct 21, 2026
Full-Page Modal Analytics	Added full-page expansion capability to drilldown report modals (`DrilldownReportModal.tsx`) and interactive chart tables (`InteractiveChartTableModal.tsx`) with responsive viewport adaptation.	Enables GM and senior leadership to inspect complex multi-tier cross-tabulations without clipping.	**PRODUCTION** Oct 24, 2026
Near-Expiry Slicing & Power SKUs	Excluded Near-Expiry clearance outlets from regular Power SKU compliance scoring; live canonical serving (MT: 24, TT: 27) with dynamic flavor consolidation (Scotch Vanilla TT).	Prevents unfair skew of brand distribution scores caused by discount clearance channels.	**PRODUCTION** Oct 24, 2026
Cold Chain Visual Alignment	Harmonized visual status gauges and underlying mathematical calculations for refrigeration units, strictly enforcing 3% critical breach and 97% within-range thresholds.	Ensures total visual consistency between executive dashboard status and physical cooler audits.	**PRODUCTION** Oct 21, 2026
Master Sync Pipeline	Enterprise admin pipeline to upload, parse, and persist updated Excel master files (Power SKUs, Classification) directly to server disk with zero container rebuilds and automated cache eviction.	Empowers commercial admin to update product catalogs & customer tiers without developer redeployments.	**PRODUCTION** Oct 18, 2026
Frequency Target Tracking	Real-time tracking of supervisor visit frequency against monthly classification targets (Class A: 4/mo, Class B: 2/mo, Class C: 1/mo, Class D: 1 every 2 mos) with executive compliance guidelines banner.	Enforces operational route discipline; reveals under-visited key accounts before revenue impacts occur.	**PRODUCTION** Oct 15, 2026
GM Directives Workflow	Executive leadership can flag specific audit photos with corrective directives (Assigned Supervisor, Action Required, Priority). Supervisors must upload camera-only photo proof to mark resolved.	Closed-loop accountability: GM directives are verified with visual proof before closure.	**PRODUCTION** Sep 24, 2026
High-Res Lightbox Dock	Interactive full-screen inspection modal with bottom dock controls: smooth zoom (up to 400%), mouse wheel zoom, 90° incremental rotation, panning, and observation details drawer.	Allows GM and management to scrutinize shelf planograms, share of voice, and asset condition.	**PRODUCTION** Oct 10, 2026
Multi-Asset Tracking	Comprehensive tracking for outlets with multiple refrigeration units (Chiller 1, Chiller 2, Freezer 1, Freezer 2) with exact size model identification and temperature compliance.	Full visibility into multi-cooler stores and cold-chain equipment health.	**PRODUCTION** Oct 10, 2026
Offline PWA Engine	Full Progressive Web App with custom Service Worker (`sw.ts`), client data compression, image down-sampling, and offline IndexedDB background queue sync.	Supervisors audit remote outlets without cellular connectivity; zero data loss.	**PRODUCTION** Aug 31, 2026
Custom Domain & Origin Retention	Azure proxy header normalization (`x-forwarded-host`), preventing unwanted azurewebsites.net redirects, ensuring seamless corporate domain experience on `dandyapp.tech`.	Maintains branded identity and eliminates corporate session invalidations across cloud proxies.	**PRODUCTION** Oct 22, 2026
Fit-to-Screen & All Export	Comprehensive Visit Execution Table (`VisitSummaryTable.tsx`) with toggleable "Fit to Screen (No Scroll)" mode, quick modal inspect, column sorting, and unpaginated "All" records Excel export.	Executive dashboard view fits on standard laptop screens without awkward horizontal scrolling.	**PRODUCTION** Oct 14, 2026

3. Critical Architectural Bug Fixes & Problem Remediations

Issue Identified	Technical Root Cause & Remediation Implemented	Operational Impact	Resolved
MSSQL Deadlock 1205 & Timeout 499	High concurrent customer/route master bulk ingestion caused lock escalation and deadlock collisions. *Remediated by implementing derived table chunking, table hints `WITH (HOLDLOCK)`, and exponential backoff retry wrappers in `lib/db.ts`.*	Eliminates ingestion crashes and enables simultaneous supervisor visit submissions during bulk syncs.	RESOLVED Oct 22, 2026
MSSQL Parameter `@p0` Collisions	Multi-statement transactions re-used identical request contexts, triggering `@p0 already declared` exceptions during visit submissions. *Remediated by instantiating fresh `new sql.Request(transaction)` per statement.*	Guarantees 100% reliable visit submission transactions on Microsoft Azure SQL Server.	RESOLVED Oct 22–24, 2026
Canonical Auth & Supervisor Drift	Supervisor credentials between local environments and `dandyapp.tech` exhibited schema divergence and Server Action timeout failures. *Remediated by automated canonical synchronization (`sync-canonical-users.js`) and REST API routing.*	Complete credential parity across all 20 supervisors with fast, dependable administrative updates.	RESOLVED Oct 24, 2026
Dairy Denominator Distortion	Classification tier coverage percentages used outdated baseline constants. *Remediated by calibrating baseline universe to verified 2,535 total outlets (Class A: 341, Class B: 551, Class C: 648, Class D: 434, Class E: 561).*	Restores statistical precision to executive market coverage and route penetration metrics.	RESOLVED Oct 21, 2026
Cold Chain Visual State Conflict	Visual status indicators showed amber alerts while raw data indicated compliant operation. *Remediated by strictly aligning frontend calculation thresholds to 97% within-range and 3% breach standards.*	Eliminates false alarms and aligns visual KPI presentation with physical equipment audits.	RESOLVED Oct 21, 2026
Azure Host Redirection Leaks	Next-Auth redirects bypassed the canonical `dandyapp.tech` domain, sending users to default Azure hostnames. *Remediated by sanitizing incoming `x-forwarded-host` headers and dynamically enforcing origin in `auth.config.ts`.*	Preserves corporate branding and prevents SSL/cross-origin authentication drops.	RESOLVED Oct 20–22, 2026
Action Comments Lost on Deploy	Container redeployments or visit re-syncs risked overwriting supervisor action comments. *Remediated by implementing native T-SQL MERGE (`MERGE INTO ActionComments ... WHEN MATCHED THEN UPDATE`) dual-engine logic.*	Permanent data preservation of all audit remarks and directive action comments on Azure SQL.	RESOLVED Oct 18, 2026
Stale PWA Offline Interception	Service worker inadvertently cached Next.js API mutations in offline mode, causing stale responses. *Remediated by strictly excluding `/api/*` and Server Action routes from Workbox offline fallback caches.*	Guarantees live dynamic server data handoff immediately when cellular or Wi-Fi reconnects.	RESOLVED Oct 17, 2026
Master File Ephemeral Loss	Uploaded master Excel files were stored in ephemeral runtime memory, losing updates on container recycling. *Remediated by engineering persistent local disk storage in `data/` directory with automated boot re-hydration.*	Master updates survive server reboots and container redeployments indefinitely.	RESOLVED Oct 15, 2026
Directive Action Leak	Flagged directives in Audit Photo Gallery were displaying across all photos and supervisors. *Remediated by strictly isolating directives to compound key (`photoId + outletCode + supervisor`).*	Directives now appear exclusively on the exact photo, outlet, and supervisor intended.	RESOLVED Oct 14, 2026
Power SKU Denominator	Power SKU compliance was defaulting to 41 SKUs or exceeding 100%. *Remediated by strictly enforcing business rule: MT = 24 Power SKUs, TT = 27 Power SKUs, with Y-axis strictly capped at 100%.*	Audited scores represent true store compliance without mathematical distortion.	RESOLVED Oct 11, 2026
Saifullah vs Saif Ambiguity	Two supervisors with similar names were colliding in reports. *Remediated by strictly separating Supervisor Saifullah (Manager: Adnan) from Supervisor Saif (Manager: Ashfaq) in database queries and slicers.*	Completely eradicated supervisor misattribution and reporting cross-talk.	RESOLVED Sep 30, 2026
Master Data Import Crash	Server Action threw foreign key constraint errors when re-importing customer master files. *Remediated by implementing ordered upsert transactions that preserve active audit linkages.*	Weekly customer master updates run seamlessly without wiping audit history.	RESOLVED Oct 13, 2026
Azure Deployment Mismatch	Next.js standard build failed on Azure App Service due to missing runtime dependencies. *Remediated by switching to standalone container output (`output: 'standalone'`) with runtime env injection.*	Permanent deployment reliability on Microsoft Azure cloud infrastructure.	RESOLVED Sep 15, 2026

4. Core Technical Architecture & Security Model

Enterprise Architecture Overview

The platform is engineered on a resilient multi-tier enterprise stack designed for high throughput, sub-second query response times, and resilience in low-bandwidth field environments:

Layer / Subsystem	Technical Implementation & Enterprise Governance
Frontend & Client Layer	Next.js 16 (App Router), React 19, TypeScript, Vanilla CSS design tokens with executive dark/light themes, Lucide icon system, and responsive viewport fitting without horizontal scroll.
Offline & Edge PWA Engine	Custom Service Worker (`sw.ts`) using Serwist/Workbox, smart route bypass for dynamic Next.js `/api/*` mutations, client-side JPEG down-sampling (`image-compressor.ts`), and IndexedDB background sync queue.
High-Concurrency MSSQL Engine	Azure SQL Server with dedicated connection pooling, isolated `sql.Request` transaction scopes, derived table batching, and `WITH (HOLDLOCK)` table hints eliminating Deadlock 1205.
Dual-Engine Database Failover	Dual-Engine Architecture natively supporting Microsoft Azure SQL Server (production) with automated failover detection to Railway/MySQL, complete with engine-specific T-SQL vs standard SQL syntax handlers.
Master Data Pipeline & Cache	Disk-persisted runtime Excel ingestion in `data/` directory, dynamic schema validation, automated in-memory cache eviction (`dashboard-cache.ts`), and real-time live synchronization across field devices.
Cloud Hosting & DevOps CI/CD	Microsoft Azure App Service Linux container hosting, Next.js standalone container deployment, custom domain binding to `dandyapp.tech`, and automated GitHub Actions CI/CD pipeline.
Identity & Data Security Model	Role-Based Access Control (Admin, Sub-Admin, Supervisor), bcrypt password hashing, NextAuth v5 session hardening with proxy host sanitization, and tamper-resistant camera-only proof enforcement.

5. Conclusion & Operational Readiness

The Field Visit Management Platform has evolved from an initial mobile prototype into a hardened, enterprise-grade audit operating system. Across **286 engineering hours, 164 production commits, and 9 continuous weeks of delivery**, every operational directive specified by commercial management and executive leadership has been rigorously built, tested, and validated under peak concurrency.

The system is currently **100% active, stable, and production-ready (v3.0)** on `dandyapp.tech`, empowering the General Manager and executive leadership with real-time operational visibility into store execution, asset preservation, and field force accountability.

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